Birla Institute of Technology and Sciences, Pilani, Goa Campus
Machine Learning for Electronics Engineers
EEE G513

Test 1 (15%)

Max Time: 75 minutes
Max Marks: 45

Instructions:
1. In case of MCQs, **one or more answers** may be correct. Choose the right answers with an '**X**'
2. In case of numerical questions, write the answer correct to **two decimal places**.
3. In case of descriptive answers, encircle your final answers. Step wise marks will be given.

Name:
ID Number:

Q1. (4 marks) The probability that an archer hits her target when it is windy is 0.4; when it is not windy, her probability of hitting the target is 0.7. On any shot, the probability of a gust of wind is 0.3. Find the probability that

(i) on a given shot there is a gust of wind and she hits her target.	
(ii) she hits the target with her first shot.	
(iii) she hits the target exactly once in two shots.	
(iv) on an occasion when she missed, there was no gust of wind.	

Q2. (2 marks) An archery target is made of 3 concentric circles of radii $\frac{1}{\sqrt{3}}$, 1 and $\sqrt{3}$ feet. Arrows striking within the inner circle are awarded 4 points, arrows within the middle ring are awarded 3 points, and arrows within the outer ring are awarded 2 points. Shots outside the target are awarded 0 points.

Consider a random variable X, the distance of the strike from the center in feet, and let the probability density function of X be

$$f(x) = \begin{cases} \frac{2}{\pi(1+x^2)} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

What is the expected value of the score of a single strike?

A) 2.17		C) 4.16	
B) 2.83		D) 3.23	

Q3. (2 marks) Ordinary least-squares regression is equivalent to assuming that each data point is generated according to a linear function of the input plus zero-mean, constant-variance Gaussian noise. In many systems, however, the noise variance is itself a positive linear function of the input (which is assumed to be non-negative, i.e., x ≥ 0). Which of the following families of probability models correctly describes this situation in the univariate case?

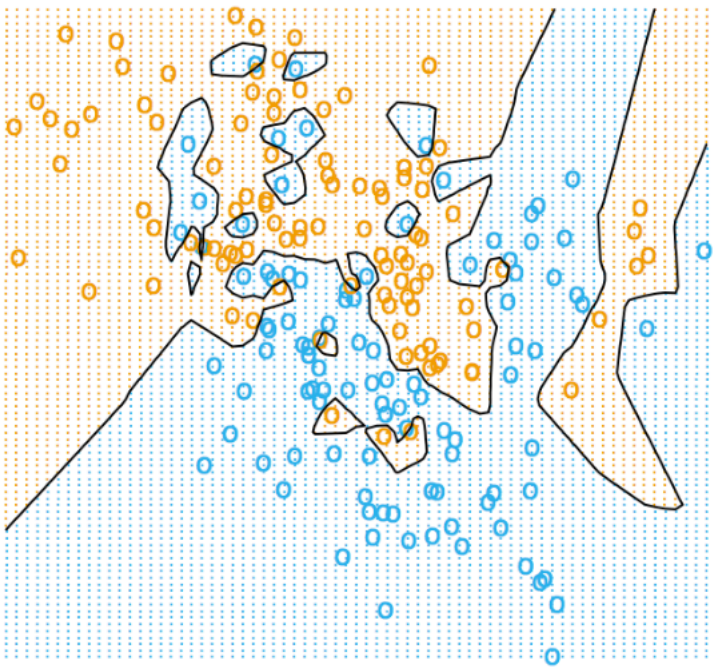
A) $P(y\|x) = \frac{1}{\sigma\sqrt{2\pi x}}\exp(-\frac{(y-(w_0+w_1x))^2}{2x\sigma^2})$		C) $P(y\|x) = \frac{1}{\sigma\sqrt{2\pi x}}\exp(-\frac{(y-(w_0+(w_1+\sigma^2)x))^2}{2\sigma^2})$	
B) $P(y\|x) = \frac{1}{\sigma\sqrt{2\pi}}\exp(-\frac{(y-(w_0+w_1x))^2}{2\sigma^2})$		D) $P(y\|x) = \frac{1}{\sigma x\sqrt{2\pi}}\exp(-\frac{(y-(w_0+w_1x))^2}{2x^2\sigma^2})$	

Q4. (4 marks) We have constructed the multiple choice problems such that every false positive will incur some negative penalty. For one of these multiple choice problems, given that there are p points, r correct answers, and k choices, what is the formula for the penalty such that the expected value of random guessing is equal to 0? (You may assume k > r)

Q5. (2 marks) You fit a logistic regression model on spam data for a Kaggle Competition. Assume you had a very good score on the public test set, but when the examiners ran your model on a private test set, your score dropped a lot. This is likely because you overfitted by submitting multiple times and changing the following between submissions:

A) λ, your penalty term		C) ε, your convergence criterion	
B) η, your step size		D) Fixing a random bug	

Q6. (2 marks) Which of the these classifiers could have generated this decision boundary?

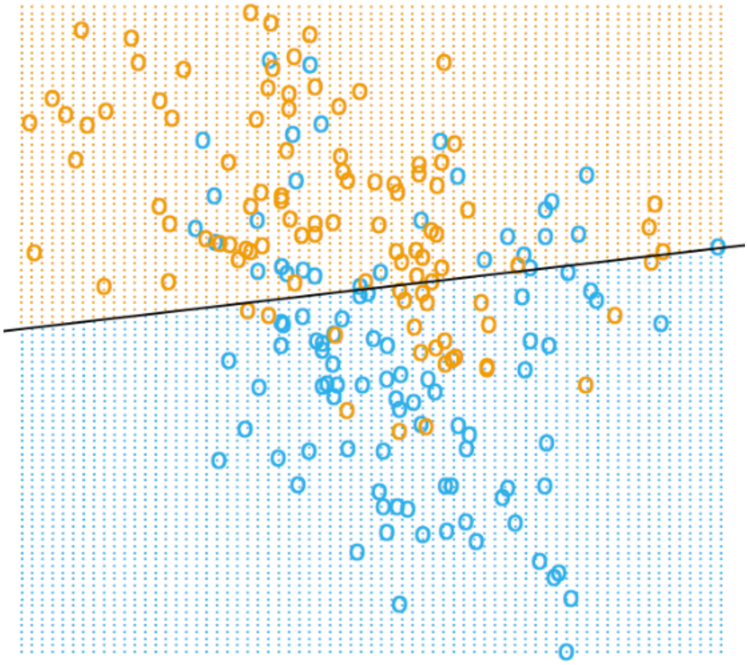


A) Naive Bayes		C) 1-NN	
B) Logistic Regression		D) None of the Above	

Q6. (2 marks) Suppose that we want to estimate the ideal parameter θ^* for $p(x, y, \theta)$ given a set of observations $\{x_i, y_i\}$. Which of the following is a key assumption made when using $\theta^{MLE} = \text{argmax}_\theta \Sigma_i log(p(x_i, y_i | \theta_i))$ for Maximum Likelihood Estimation (MLE) to estimate the model parameter?

A) The data is normally distributed.		B) The data is independent and identically distributed (i.i.d.).	
B) The data contains no outliers.		D) The data is linearly separable.	

Q7. (2 marks) Which of the these classifiers could have generated this decision boundary?



A) Logistic Regression		C) 2-NN	
B) Naive Bayes		D) Perceptron	

Q8. (2 marks) We are drawing sample points from a distribution with the probability density function (PDF) $f(x) = 0.5 \cdot e^{-|X-\mu|}$, but we do not know the mean $\mu \in \mathscr{R}$. We decide to estimate μ with maximum likelihood estimation (MLE). Unfortunately, we have only two sample points $X1, X2 \in \mathscr{R}$. Assuming that $X_1 \leq \mu \leq X_2$ then the log-likelihood:

A) Increases with increasing μ		C) Exponentially Increases with increasing μ	
B) Decreases with increasing μ		D) is independent of μ	

Q9. (5 marks) The table below is a list of sample points in R^2. Suppose that we run the perceptron algorithm, with a fictitious dimension, on these sample points.

x_1	x_2	y
-3	2	$+1$
-1	1	$+1$
-1	-1	-1
2	2	-1
1	-1	-1

Suppose that the learning rate is $\epsilon = 1$ and the initial weight vector $w^0 = (-3,2,1)$, where the last component is the bias term. What is the equation of the separating line found by the algorithm, in terms of the features x_1 and x_2?

Ans.

Q10. (18 marks) In lecture, we saw how least-squares regression is motivated by maximum likelihood estimation if we think our data obeys a linear relationship but has added noise that is normally distributed. But what if the noise is better modeled by the Laplace distribution? The PDF of the Laplace distribution with mean μ and scale parameter β is defined as

$$f(x \mid \mu, \beta) = \frac{1}{2\beta} e^{\frac{|x-\mu|}{b}}$$

Following our customary notation, the input is an $n \times d$ design matrix X and a vector y such that y_i is the label for sample point X_i, where $X_i^\top$ is the i^{th} row of X. To keep things simple, we will do linear regression through the origin (no bias term w_0), so the regression function is $h(x) = w \cdot x$. Our model is that each label y_i comes from a linear relationship perturbed by Laplacian noise,

$$y_i \sim Laplace(w \cdot X_i, \beta)$$

where $w \in \mathrm{R}^d$ is the true linear relationship. We will use maximum likelihood estimation to try to estimate w.

(a) [4 marks] Write the likelihood function $L(w; X, y)$ for the parameter w, given the fixed data X and y.

(b) [3 marks] Write the log likelihood function $l(w; X, y)$ for the parameter w, given the fixed data X and y, in as simple a form as you can. (Make sure your logarithms are written in natural base.)

(c) [3 marks] What is the simplest cost function we can minimize that gives us the same value of *w*as maximizing the likelihood?

(d) [4 marks] How is the cost function you just derived different from standard least-squares regression? Is it more or less sensitive to outliers? Why? Justify.

(e) [4 marks] Write the batch gradient descent rule for minimizing your cost function, using η for the step size (aka learning rate). You may omit training points whose losses have undefined gradients. *Hint:* Recall that $\frac{d|\alpha|}{d\alpha}$ is 1 for $\alpha > 0$, -1 for $\alpha < 0$, and undefined for $\alpha = 0$.

Space for Rough Work